THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The formula of the compound formed between Mg and Cl is: 1

(a) Mg_2Cl_2 **(b)** MgCl **(c)** $MgCl_2$ **(d)** Mg_2Cl

2. Which is NOT a method of preventing corrosion? 1

(a) alloying **(b)** heating in air **(c)** galvanising **(d)** painting

3. Which gas is not evolved when metals react with dilute nitric acid (except Mg, Mn)? 1

(a) nitrogen dioxide **(b)** hydrogen **(c)** nitric oxide **(d)** nitrous oxide

4. The compound formed by Na and O has the formula: 1

(a) NaO **(b)** NaO_2 **(c)** Na_2O **(d)** Na_2O_2

5. The iron pillar at Delhi has not rusted for centuries mainly because: 1

(a) it is kept in oil
(b) it is made of pure gold
(c) a thin protective layer forms on its surface
(d) it is painted every year

6. The metal which can be extracted by heating its sulphide ore in air is: 1

(a) iron **(b)** sodium **(c)** aluminium **(d)** mercury

7. Ionic compounds are brittle because: 1

(a) they are non-polar
(b) ions are loosely bound
(c) they are soft
(d) like charges come close on shifting layers

8. The cathode in electrolytic refining of copper is: 1

(a) graphite **(b)** zinc
(c) impure copper **(d)** a thin strip of pure copper

9. Metals of medium reactivity are mostly found as: 1

(a) chlorides only **(b)** free state
(c) nitrates **(d)** sulphides and carbonates

10. Which compound conducts electricity in the molten state? 1

(a) NaCl **(b)** $C_6H_{12}O_6$ **(c)** sugar **(d)** CCl_4

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between roasting and calcination. 2

12. Why does an iron nail kept in copper sulphate solution turn brownish? 2

13. Why is carbon not used to extract sodium from its oxide? 2

14. Why is sodium obtained by electrolysis and not by heating its ore? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. An element X has atomic number 12 and Y has atomic number 17. Predict the formula of the compound and the type of bond. 3

16. Explain why alloying is done. How does the property of iron change by adding carbon, nickel and chromium? 3

17. What are amphoteric oxides? Write two examples with the reaction of one with acid. 3

OR

Explain the reaction of metals with dilute acids with Mg, Zn and Cu. 3

18. Explain with equations: 3
(i) iron with steam
(ii) calcium with water.

OR

Explain why copper vessels get a green coating and silver turns black. Name the compounds. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A metal M forms an oxide MO, soluble in acids and bases. Identify the nature of M and write the reactions. 5

OR

Explain the reducing action of aluminium in the thermit reaction and state why it is exothermic. Write the reaction. 5

20. Describe the reaction of Na, Ca, Mg, Al, Fe with water and give the reason for the differences. 5

OR

Compare ionic and covalent compounds on melting point, conductivity, solubility in water, and structure, giving examples. 5

END OF QUESTION PAPER — Total: 40 Marks

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